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import java.io.*;

class numberAnalysis{

void prime(int n){
if(n <= 1 ){
System.out.println("Not Prime");
return;
}
for(int i=2; i<=n-1; i++){
if(n%i == 0){
System.out.println("Not Prime number");
return;
}
}
System.out.println("Prime number");

}

void palindrome(int n){

int original = n;
int rev=0;
int digit;

while(n!=0){
digit = n%10;
rev = rev*10 + digit;
n = n/10;
}

if(original == rev){
System.out.println("Palindrome number");
} else {
System.out.println("not a Palindrome number");
}
}

void armstrong (int n){
int original = n;
int count = 0;
int sum = 0;

while(n!=0){
count++;
n = n/10;
}

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n = original;

while(n!=0){
int num = n%10;
sum = sum + (int)Math.pow(num,count);
n = n/10;
}

if(original == sum){
System.out.println("Armstrong number");
} else {
System.out.println("Not an Armstrong number");
}
}

void sumOfDig(int n){
int sum = 0;

while(n!=0){
int digit = n%10;
sum = sum + digit;
n = n/10;
}
System.out.println(sum);
}

void reverseNum(int n){
int rev = 0;

while(n!=0){
int digit = n%10;
rev = rev*10 + digit;
n = n/10;
}
System.out.println(rev);
}

public static void main(String args[]) throws IOException{
BufferedReader br = new BufferedReader(new InputStreamReader(System.in));

OperationMenu ob = new OperationMenu();

System.out.println("Enter a number: ");
int n = Integer.parseInt(br.readLine());

if(n < 0) {
System.out.println("Please enter a positive number.");
}
}

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return;  
}
```

```
int option;  
do {  
    System.out.println("MENU: ");  
    System.out.println("1. Check whether the number is Prime");  
    System.out.println("2. Check whether it is Palindrome");  
    System.out.println("3. Check whether it is Armstrong");  
    System.out.println("4. Find the Sum of Digits");  
    System.out.println("5. Reverse the Number");  
    System.out.println("6. Exit");
```

```
    System.out.println("Enter your option: ");
```

```
    option = Integer.parseInt(br.readLine());
```

```
    switch(option){  
        case 1:  
            ob.prime(n);  
            break;
```

```
        case 2:  
            ob.palindrome(n);  
            break;
```

```
        case 3:  
            ob.armstrong(n);  
            break;
```

```
        case 4:  
            ob.sumOfDig(n);  
            break;
```

```
        case 5:  
            ob.reverseNum(n);  
            break;
```

```
        case 6:  
            System.out.println("Exited");  
            break;
```

```
        default:  
            System.out.println("Invalid Option");
```

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    }  
} while(option!=6);  
}
```

